

Week 6 Cross-Track Integration Record — Generative AI ↔ Data Science

GenAI track representative: [Your Name] **Data Science track counterpart:** [Name / track lead]

Integration type: Tool / interface integration **Status:** Complete — merged and verified

1. Track collaborated with

Data Science track (no-show prediction).

Secondary coordination with **Project Management** for issue/risk logging.

2. Project dependency

The GenAI assistant needed a way to answer user questions such as *"What is the risk of this appointment being missed?"* with a real prediction rather than a generic KB answer.

The Data Science track owns the no-show classification model and the associated feature schema. The GenAI track owns the assistant, the conversational routing, and the response format presented to the patient.

3. Information / output received

From Data Science:

- **Feature schema** for the no-show model:
 - `lead_time_days` — days between booking and appointment
 - `appointment_hour` — hour of day (0–23)
 - `age` — patient age
 - `prior_no_shows` — count of previous no-shows
 - `sms_received` — boolean reminder flag
- **Prediction contract:** probability in `[0, 1]`; classified as `high` (≥ 0.30), `medium` (≥ 0.15), `low`

(otherwise).

- **Feature importance insight** (from Data Analytics via Data Science): long lead time, prior no-shows, and absence of SMS reminder are the most influential signals.
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4. Information / output provided

To Data Science:

- **Tool interface wrapper** — `app/tools/no_show_predictor.py`, with a stable `run(**features) -> dict` interface.
 - **Heuristic fallback** — a documented, feature-aligned heuristic used when the trained model artifact is unavailable. This ensures the GenAI pipeline is never blocked by a missing DS artifact and gives Data Science a well-defined target for the eventual artifact path (`models/no_show_model.pkl`).
 - **Documented response format** — probability, risk band, top drivers, source (model vs heuristic).
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5. Integration activity completed

1. Agreed on the feature schema and prediction contract.
 2. Implemented `app/tools/no_show_predictor.py` in the GenAI repository.
 3. Wired the tool into the assistant's `no_show_risk` intent path in `app/agents/core_agents/knowledge_agent.py`.
 4. Verified end-to-end that a `no_show_risk` query reaches the tool and returns a formatted response with probability, risk band, and drivers.
 5. Added the tool to the Week 6 evaluation suite (DS-01, DS-02).
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6. What changed as a result

Before Week 6:

- The assistant had no `no_show_risk` capability. A user asking about no-show risk received a generic KB answer.
- Data Science's model was not reachable from the assistant layer.

After Week 6:

- The assistant recognises the `no_show_risk` intent and calls `NoShowPredictorTool`.
- Responses include probability, risk band, and top drivers, using the DS-provided feature schema.
- If the DS artifact is not present, the assistant still returns a documented heuristic so the capability is not blocked.

Evaluation results:

- `DS-01` — "Why do patients miss appointments at HealthConnect?" — PASS
- `DS-02` — "What can HealthConnect do to reduce no-shows?" — PASS

7. Evidence

Artefact	Location
No-show tool implementation	<code>app/tools/no_show_predictor.py</code>
Assistant integration	<code>app/agents/core_agents/knowledge_agent.py</code>
Evaluation scenarios DS-01, DS-02	<code>tests/fixtures/assistant_eval_set.json</code>
Evaluation results (after)	<code>docs/week6/evidence/w6_eval_after.json</code>
Before/after evidence	<code>docs/week6/evidence/w6_before_after_examples.json</code>
Git commit	<code>Week 6 (GenAI): assistant evaluation & refinement</code>

8. Notes on coordination

This integration was not only a communication exchange. It produced:

- A new file in the GenAI repository (`no_show_predictor.py`)
- A new intent path in the assistant (`no_show_risk`)
- Two new evaluation scenarios that exercise the integration
- A documented interface contract that Data Science can target when the trained artifact is ready

The integration is therefore "meaningful" per the Week 6 assignment definition — it produced a new artefact, a code change, and a validated behaviour.

